

Track-Aligned Interaction Modeling for Group Activity Recognition

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34th Signal Processing and Communications Applications Conference (SIU 2026)

Group Activity Recognition · Dynamic Graphs · Track-Aligned Supervision

Problem: group labels depend on relations, not isolated actions

GROUP ACTIVITY RECOGNITION

Given a short multi-person clip, predict the collective activity.

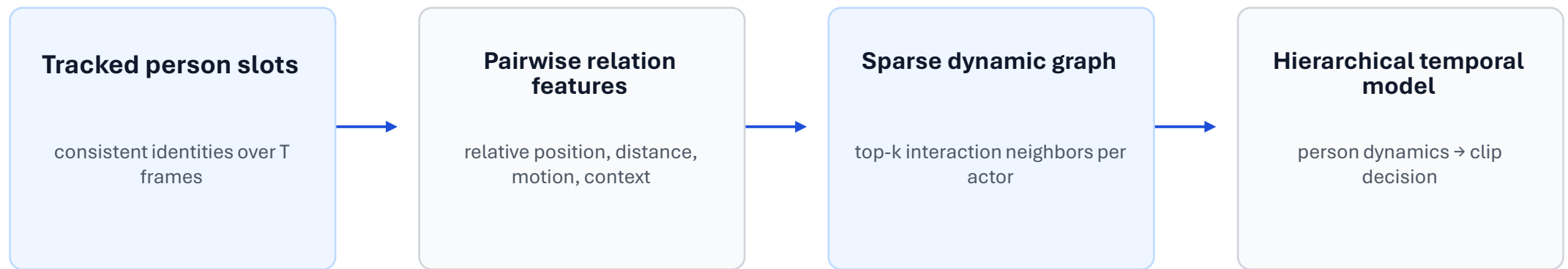
- The important evidence is distributed across people and time.
- Only a subset of actors may determine the final label.
- The model must track who interacts with whom.

What makes GAR hard?

- **Identity consistency**
same person across frames
- **Interaction structure**
relevant neighbors only
- **Temporal supervision**
roles should be stable over time

We model tracked people as a dynamic interaction graph

Instead of treating actor annotations independently, the paper represents each clip as temporally aligned person trajectories and performs interaction reasoning over those trajectories.



Input representation: person-centric and track-aligned

Each clip becomes a fixed $T \times N$ tensor of person slots.

- Pose coordinates and pose motion
- Bounding-box geometry and box motion
- Context cues such as side or object-relative position
- Appearance features

$$\mathbf{x}_i^t = [\text{pose}, \Delta\text{pose}, \text{box}, \Delta\text{box}, \text{context}, \text{slot}]$$

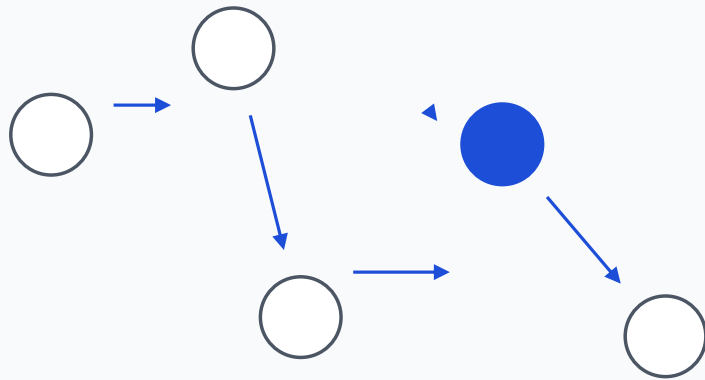
Pairwise relations are explicit, not left entirely to attention.

- Relative position
- Distance
- Relative motion
- Contextual relations

$$\mathbf{r}_{ij}^t = [\Delta\mathbf{x}_{ij}^t, d_{ij}^t, \Delta\mathbf{v}_{ij}^t, \text{context}_{ij}^t]$$

Dynamic sparse interaction graph

For each receiver actor, the model scores all other actors and keeps only the top-k neighbors.



Directed top-k graph

Scoring

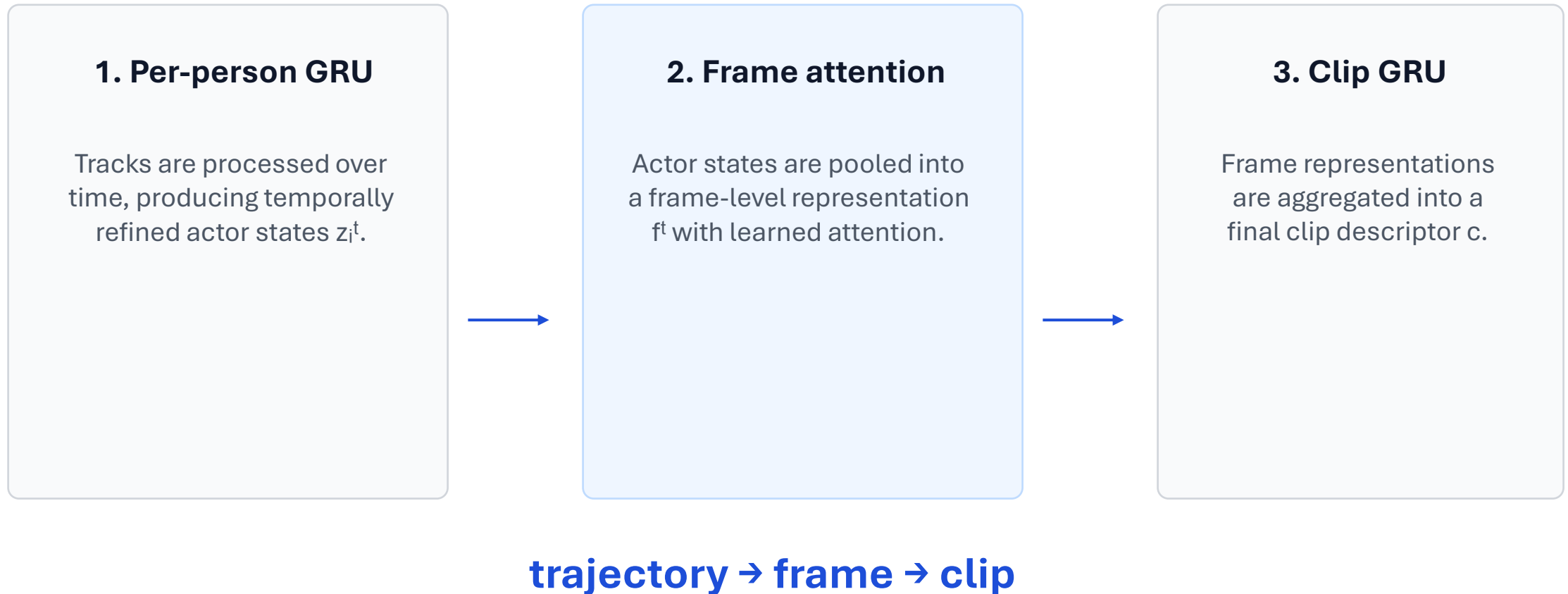
$$\alpha_{ij}^t = \text{softmax}_j f([h_i^t, h_j^t, r_{ij}^t])$$

Message passing

$$m_i^t = \sum_{j \in N(i)} \alpha_{ij}^t \cdot g([h_j^t, r_{ij}^t])$$

Two stacked interaction stages refine person context beyond one-hop relations.

Hierarchical temporal reasoning



Track-aligned sequence supervision

The auxiliary action loss is applied along person trajectories instead of only at a single frame.

Common auxiliary supervision



only center frame is labeled

Track-aligned supervision



labels follow the same person through time

$$L = L_{\text{group}} + \lambda L_{\text{action}}$$

Experimental setup

Datasets

- Volleyball: team activities in short clips
- Collective: pedestrian group activities in surveillance-style scenes
- Metric: clip-level group accuracy

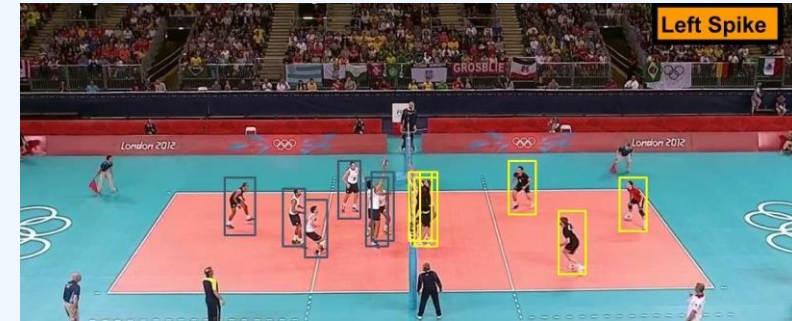
94.1%

Volleyball accuracy

95.7%

Collective accuracy

Dataset examples



Volleyball dataset



Collective dataset

Main results: best accuracy on both benchmarks

Method	Volleyball ↑	Collective ↑
PCTDM	90.3	92.2
stagNet	89.3	89.1
ARG	91.1	92.3
PRL	91.4	93.8
DIN	93.1	95.3
HiGCIN	91.4	93.0
ST-KI	93.7	95.6
Ours	94.1	95.7

94.1%

Volleyball group accuracy

95.7%

Collective group accuracy

Takeaway: the gains are modest but consistent, and the method wins on both standard GAR benchmarks.

Ablation: each component matters

Variant	Volleyball	Collective
Full model	0.94	0.96
No sparse graph	0.93	0.90
No relation features	0.92	0.83
No temporal modeling	0.89	0.87
No auxiliary supervision	0.91	0.88

Largest drops

- No relation features: Collective drops from 0.96 to 0.83.
- No temporal modeling: Volleyball drops from 0.94 to 0.89.
- No auxiliary supervision: both datasets degrade strongly.

The result is not carried by one trick. Graph structure, relation cues, temporal modeling, and supervision all contribute.

Conclusion

Track-aligned interaction modeling improves group activity recognition by making identities, relations, and temporal supervision explicit.

- Dynamic sparse graphs focus interaction reasoning on relevant actors.
- Hierarchical GRUs capture person-level and clip-level temporal structure.
- Track-aligned action supervision stabilizes role representations.
- The method obtains 94.1% on Volleyball and 95.7% on Collective.

Thank you.

Questions?